

THESHIKA NAVOD | COMPUTER ENGINEERING

theshikanavod@gmail.com | +94713621414 | [GitHub](#) | [LinkedIn](#) | [Medium](#) | [Website](#) | Sri Lanka

PROFILE SUMMARY

Software Engineer specializing in **Java, Spring Boot, Microservices, Distributed Systems, and Cloud-native development**, with strong experience building **production-grade AI systems** including RAG pipelines, agentic architectures, and multimodal applications. Hands-on expertise in **Kafka, gRPC, CI/CD, Docker, Databases**, and scalable backend engineering. Passionate about building reliable, high-performance systems.

WORK EXPERIENCE

Information Systems Associates (ISA) Intern Software Engineer

July 2024 – January 2025

- Developed and maintained microservices for Air Arabia's aviation systems using Spring Boot and Angular
 - Refactored legacy codebase resolving 200+ critical bugs identified by SonarQube and increased unit test coverage from 10% to 80% using JUnit and Mockito, improving code maintainability by 60% and reducing production defects by 30%
 - Optimized system performance by migrating REST endpoints to gRPC protocol with event streaming and implementing real-time estimate departure time display, reducing inter-service latency by 40% and passenger wait time by 25%
 - Worked in Agile environment with daily standups, sprint planning, and retrospectives using Jira and Confluence
- Tech Stack:** Java (8, 11, 17), Spring Boot, React, Angular, PostgreSQL, Oracle DB, Redis, Kafka, gRPC, Docker, Jenkins, JUnit, Mockito, SonarQube, Git, SOAP/REST APIs, Agile environment with Jira/Confluence

PROJECT EXPERIENCE

AI Code Generator and Reviewer System - Research Project (code - upon request)

Built VS Code extension automating code generation and bug detection for legacy codebases, achieving 42% score on SWE-bench. Developed multi-agent reviewer system using fine-tuned LLMs detecting security vulnerabilities, code smells, and performance issues with human-in-the-loop validation, reducing manual review time by 60%.

Tech Stack: Python, TypeScript, LangGraph, LLM APIs, Multi-Agent system, Agentic RAG, AST, VS Code Extension API, Fine-tuning, Prompt engineering, Transformer Models, Pinecone.

BookFair Zone – Microservices Stall Reservation System

Built enterprise microservices platform for hall management, booking, real-time updatable map. JWT authentication, Kafka for event-driven communication. Implemented load balancer and API Gateway for traffic distribution.

Tech Stack: Java/Spring Boot, React, JPA/Hibernate, PostgreSQL, JWT, Kafka, Load Balancer, Api Gateway

Kafka Order Analytics Platform

Built real-time ordering system handling high-throughput message streaming with Apache Kafka, Avro serialization. Implemented fault-tolerant architecture, retry mechanisms and Dead Letter Queue. Live price aggregation in dashboard.

Tech Stack: Apache Kafka, Avro, Spring Boot, React, WebSocket, PostgreSQL

High-Security E-Voting System

Developed voting platform with end-to-end encryption and tamper-proof architecture. Implemented multi-factor authentication, JWT, token blacklisting, digital signature verification, and brute-force protection with rate limiting.

Tech Stack: Java, Spring Security, JWT, Role-based Auth, Multi Factor Auth, Email Verification, Cryptography

Sinhala Sign Language Learning Management System

Developed AI-powered LMS for hearing and speech impaired individuals, achieved 87% gesture recognition accuracy in real-time hand gestures using computer vision with MediaPipe pose estimation and deep learning models.

Tech Stack: Python, Computer Vision, OpenCV, MediaPipe, Deep Learning, TensorFlow, FastAPI, NLP

Jenkins pipeline for a Spring Boot application

Pipeline: Checkout, Test & Build, SonarQube Analysis, Quality Gate, Package, Docker Build, Deploy, and monitoring.

Tech Stack: Java 21, Spring Boot, PostgreSQL, Jenkins, Docker, SonarQube, Prometheus, Grafana, Gradle

HPC Optimization for LLM Parallel Prompt Processing

Optimized LLM prompt processing using HPC techniques achieving 3.27x speedup. Implemented and benchmarked OpenMP, MPI, and hybrid parallelization strategies across Pure Python, Pure C, and C + Python processing prompts.

Tech Stack: C, Python, OpenMP, MPI, HPC Techniques, Performance Optimization, API Integration

Multi-Tenant SaaS POS System | [Frontend](#) | [Backend](#)

Developed multi-tenant point-of-sale system with database-level tenant isolation supporting unlimited business clients. Built role-based authentication system using Clerk and implemented modern responsive UI using shadcn ui.

Tech Stack: Nestjs, React, Next.js, Node.js, Express, MongoDB, Clerk Auth, shadcn/ui, Tailwind CSS, Docker

EDUCATION

B.Sc. (Hons) Computer Engineering.

Faculty of Engineering, University of Ruhuna

GPA: 3.30

CERTIFICATES

[Java: Complete Java Core for Beginners with Exercises - 2025](#) - Andrill Partola ITER Academy, Udem

[Spring Framework for Java Developers: Practical Guide](#) - Andrii Piatakha, Learn-IT University, Udem

[AWS Essentials: A Complete Beginner's Guide](#) - Learnify IT, Udem

[CI/CD with Jenkins and Docker](#) - Gourav J. Shah, School of Devops, Udem

[Master in Software Architecture, Engineering and Development](#) - Arun Singhal B-Tech, MBA (IIM-B), Udem

[Learn Spring GraphQL](#) - Ezekiel Eromosei, Udem

[Programming Network Applications in Java- Cyber Quince](#), Udem

[Next.js With Tailwind CSS-Build a Frontend Ecommerce Project](#)- Vapa Academy, Udem

[Information Security Fundamentals](#) – Cyber Quince, Udem

TECHNICAL SKILLS

- **Programming Languages:** Java, Python, TypeScript, JavaScript
- **Backend:** Spring Boot, Spring Cloud, Node.js, NestJS, FastAPI
- **Frontend:** React, Next.js, Angular, HTML5, CSS3, Tailwind CSS
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, Neo4j, Pinecone
- **Software engineering:** Apache Kafka, RabbitMQ, Junit, Mockito, Unit Testing
- **DevOps & Cloud:** Docker, Kubernetes, Jenkins, AWS, CI/CD
- **Architecture:** Microservices System Design, RESTful APIs, gRPC, Event-Driven Systems, Server-Sent Events
- **AI/ML:** LangChain, LangGraph, RAG Systems, Multi-Agent Systems, Vector Databases, Fine-tuning

SOFT SKILLS

Communication & Presentation | Stress Management | Time Management | Team Collaboration | Leadership

INVOLVEMENT

- 3rd Year Field Representative – Computer Engineering
- IEEE INSL Competition – participation
- Leadership Development program - Maliyadeva College
- Haxtreme 2.0 - 4th place
- Former Senior Prefect – Mayurapada Central College
- Football Team Champion – Mayurapada Central

REFERENCES

Dr.Kushan Sudheera - Senior Lecturer, Faculty of Engineering, University of Ruhuna.. +94719693164, kushan@eie.ruh.ac.lk

Dr. Rajitha Udawalpola - Senior Lecturer, Faculty of Engineering, University of Ruhuna. + 94718578608, rajitha@eie.ruh.ac.lk